



**ADVANCED DATA ANALYSIS AND MACHINE LEARNING**

**NASA TURBOFAN A2:**

**OPTIMIZATION OF REMAINING USEFUL LIFE**

**ESTIMATION**

LUT University A.Y 2025 - 2026

Group: Amin Hassanzadehmoghaddam, Antonio Oliva, Nico Niemelä  
<https://github.com/aminhm/NASA-Turbofan-A2.git>

## Sommario

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## 1. Introduction

Predicting maintenance of aircraft engines is an important task in the aerospace industry. In this industry, accurate estimation of Remaining Useful Life (RUL) can reduce operational costs, prevent sudden downtimes, and enhance safety. The NASA Turbofan Jet Engine Degradation Dataset provides a benchmark for developing and testing data-driven models. The data contains multivariate time series data representing the operational history of multiple engines under varying conditions. This report focuses on the optimization of RUL estimation by applying advanced data analysis to the data. In particular, the work aims to reduce model complexity by selecting the best sensors and also keeps the predictions accurate. Partial Least Squares (PLS) regression is used as the primary modelling technique.

The report is structured as follows. First, the dataset is described, and exploratory data analysis is performed using Principal Component Analysis (PCA) to understand the variance structure and relationships among sensor measurements. Next, data pretreatment, sensor selection, and model calibration procedures are detailed. The results section presents the performance of optimized PLS models across different dataset partitions (FD001–FD004), followed by a discussion of findings, limitations, and recommendations for future improvements.

## 2. Materials and methods

The group will use Teams as a primary communication channel and GitHub will be used as a coding collaboration platform. Python was decided to be used as a tool to research the dataset.

### 2.1. Data Description

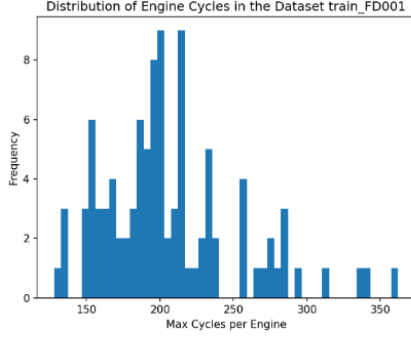


Figure 1. Distribution of engine cycles in the dataset train\_FD001.

The dataset used in this study is the NASA Turbofan Jet Engine Degradation Dataset. The dataset consists of four different datasets: FD001, FD002, FD003 and FD004. Datasets contain multiple multivariate time series, each representing the operational history of a different engine.

As expected, each engine has unequal amount of data. For example, distribution of maximum engine cycles in the dataset train\_FD001 is shown in Figure 1.

There seems to be no missing or extra values in the data.

However, datasets FD001 and FD003 have some columns with constant values. Physical meaning for operational settings and sensor measurements is not specified.

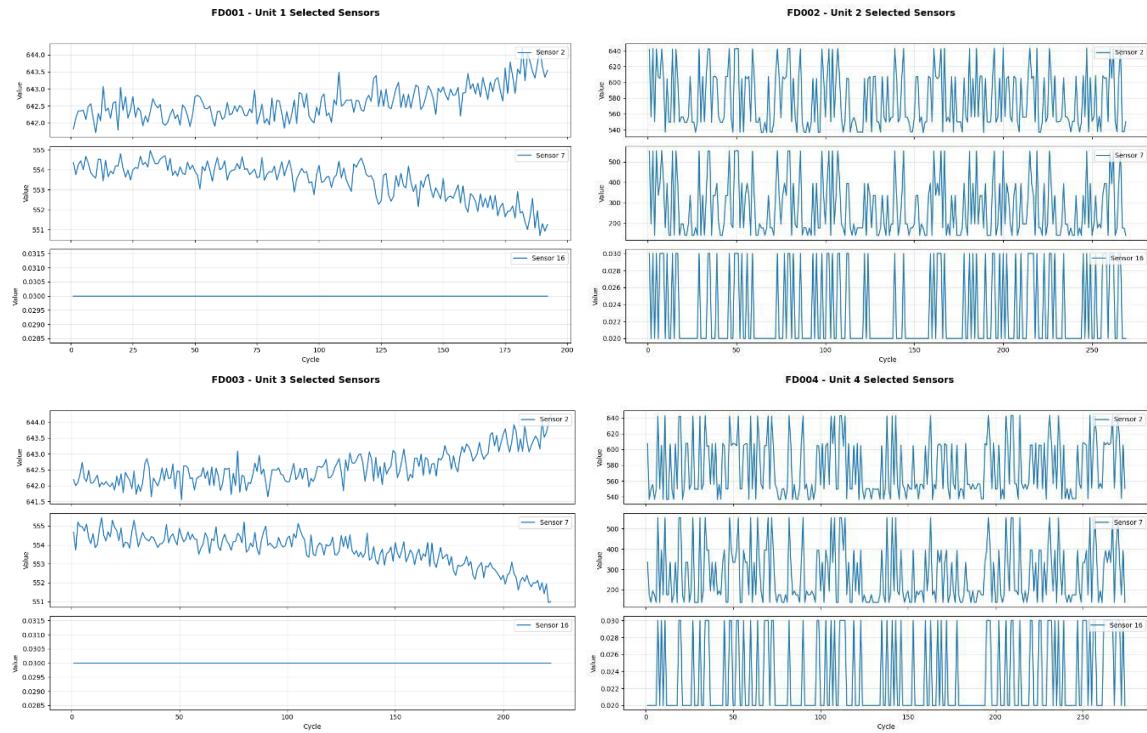


Figure 2. Distribution of variables in all datasets

The training datasets consist of 26 variables including unit number, time in cycles, three operational settings, and 21 sensor measurements. The number of observations is 20,631 for the FD001, 53,759 for the FD002, 24,720 for the FD003, and 61,249 for the FD004.

According to data, operational settings show distinct patterns. Setting 3 is nearly constant, setting 1 is normal in FD001 and FD003 but multimodal in FD002 and FD004, and setting 2 is multimodal depending on the system. Sensor measurements range from constant to normal and multimodal. They also reflect diverse engine states.

## 2.2. Exploratory data analysis with PCA

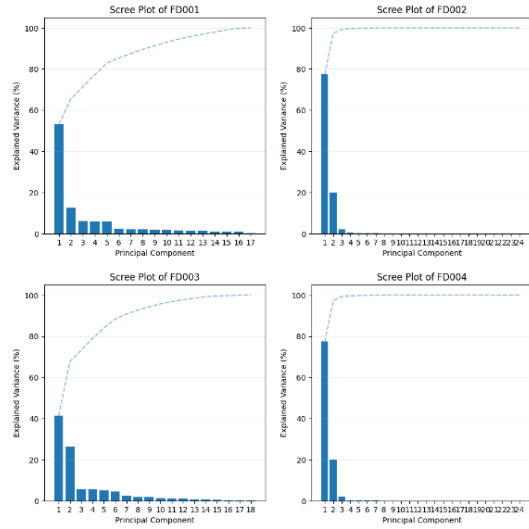


Figure 3. Scree plot of datasets with explained and cumulative variance per principal component.

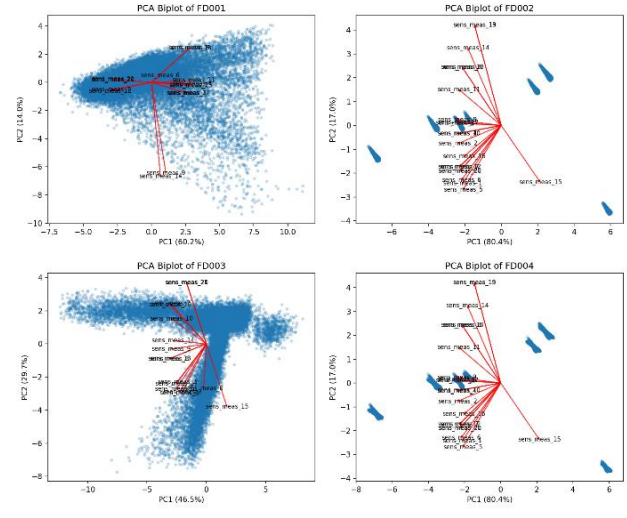


Figure 4. Biplots of data projected onto PC1 and PC2 and loadings for each sensor.

PCA was applied to the standardized matrixes of the datasets to analyse the correlations between sensors' data and retrieve the principal components that capture most of the variance in the dataset. In the scree plots in Figure 3, the first two components are the most impactful on the variance of the dataset, contributing to  $\sim 65\%$  in FD001,  $\sim 67\%$  in FD003 and  $\sim 97\%$  in both FD002 and FD004.

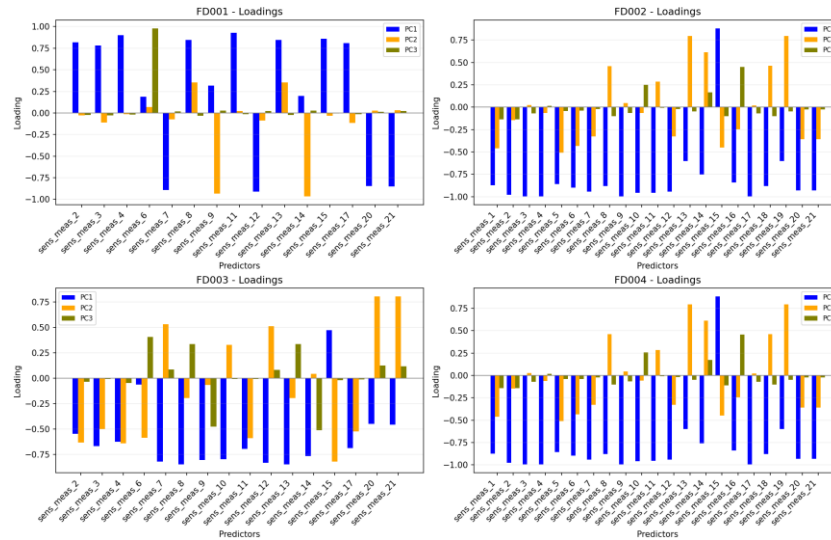


Figure 5. Loadings bar plots to analyze correlations between principal components and sensors

By looking at the plot, it is possible to determine that PC1 has a positive correlation with most sensors, while PC2 and PC3 focus attention on the differences between sensor groups. This shows that using the first three components is sufficient for further analysis.

## 2.3. Data Pretreatment

In this stage, non-informative variables were removed from each dataset. The three operational settings were excluded in all cases, as well as sensors that remain constant or near constant.

The data is centered and scaled using Standard Scaler. It subtracts the means from each sensor data, and the mean becomes zero. Then the data is divided by the standard deviation and standard deviation becomes zero. The formula of Standard Scaler is shown in Equation 1.

$$X_{standardized} = \frac{X - \mu}{\sigma} \quad (1)$$

Where  $X_{standardized}$  is standardized data,  $X$  is original data,  $\mu$  is mean of the data and  $\sigma$  is standard deviation of the data.

There are no missing values in the datasets, however, all four datasets (FD001–FD004) contain no missing values. Extreme values were found in 14 sensors for FD001–FD003 and 15 sensors for FD004, which is expected in run-to-failure engine data.

In addition, there is no misalignment in datasets which indicates that there are no missing cycles for each unit, and the original data is the same as the synchronized data.

### Visualization of pretreated data

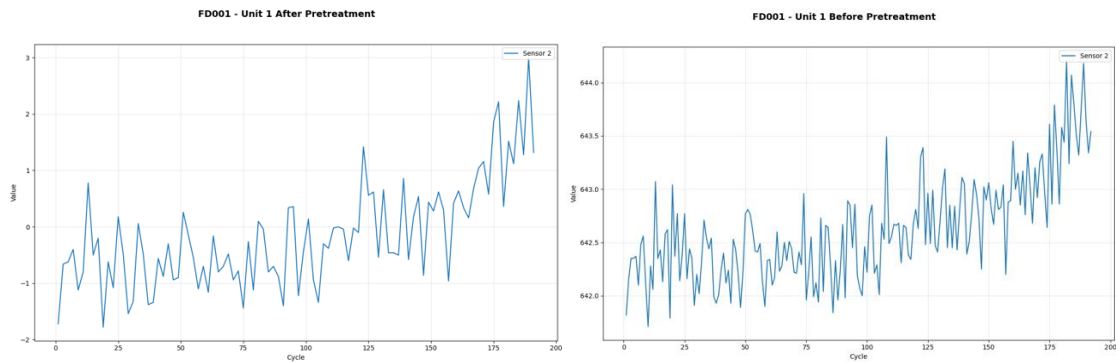


Figure 6. The data of the second sensor of FD001 dataset before and after pretreatment.

Figure 6 shows the data of the second sensor of FD001 dataset before and after pretreatment as an example.

## 2.4. Modelling Goal

The modelling goal is to recreate a regression model, removing redundant sensors from the base regression model and keeping just the most important ones, capable of calculating an estimate of the RUL of the engine calculated with all sensors.

For this purpose, four different training data were given for the analysis and recreation of the model. The best division strategy found is to split the units into three different sections:

- one for training / calibration (60% of the units),
- one for validation (20% of the units),
- one for testing (20% of the units).

## 2.5. Modelling Strategy

### Training / Calibration strategy

For training / calibration purposes, 60% of the total units are analyzed. Before the calibration step, data pretreatment was implemented to normalize data (z-score approach), remove sensors that measure constant data (not useful for the main goal) and handle missing values (not present after removing constant sensors). The main purpose of calibration is to estimate the model parameters and help identifying which sensors are more important for predicting the RUL. An expanding window cross-validation is applied to select the optimal numbers of latent variables in the PLS model, using Predicted Residual Sum of Squares (PRESS) and  $Q^2_{CV}$  as metrics. The optimal number of latent variables was chosen based on the value with maximum  $Q^2_{CV}$ .

Since, as seen in the scree plot in Figure 3, just two principal components are sufficient for describing most of the variance, it indicates that many sensors are highly correlated, and so the main strategy for this step is to use PLS, as it efficiently reduces dimensionality and handles multicollinearity among the sensor readings.

### Validation strategy

To evaluate the correctness of the parameters' estimation calculated in the training part, a validation analysis is performed using a separate set of units (20% of the total units) as a validation set.

The calibrated PLS model is evaluated on this validation set to assess its predictive performance in terms of RMSE, PRESS, and  $Q^2_{VAL}$ . The validation output is then used as feedback to assess the correctness of the model.

### Testing strategy

After a suitable subset of sensors has been found, the testing part begins. The remaining 20% of the total dataset is used to provide an unbiased estimate of predictive performance.

To assess the predictive accuracy of the computation, metrics such as RMSE and  $R^2$  are computed to ensure that the sensors subset is sufficient for estimating the exact RUL calculated using all sensors, to address any trade-off between sensor reduction and prediction quality. Also,  $Q^2_{test}$  is useful with  $R^2$  of the calibration to identify if the model is overfitted (low  $Q^2_{test}$  with high  $R^2$ ).

**Modelling flow chart**

In this case the starting point of the modelling process was when we acquired the cleaned data. In total, flow chart has 8 steps including:

- Cleaned data acquired
- Data exploration (descriptive statistics, visualizations)
- Model object definition
- Splitting the data
- Data standardization (centering, scaling, etc.)
- Initial modeling (choosing modeling algorithms, splitting data to training and validation sets, defining performance metrics)
- Model optimization (variable reduction and selection)
- Model testing (applying optimized model to the test dataset, validating results)
- Reporting the results of the analysis



## 2.7. Mathematical methods used in modelling

The group will be using PLS regression as a model to solve the task.

According to the Handbook of Partial Least Squares, PLS is a statistical method used to model complex relationships between multiple observed and latent variables. It is a component-based approach, where causality is expressed through linear conditional expectations (Esposito Vinzi et al., 2022; Wold, 1975).

The development of a PLS model begins with a training dataset consisting of  $N$  observations (such as cases, objects, compounds, or time points). Each observation includes  $K$  independent variables, denoted as  $\mathbf{x}_k$  (where  $k = 1, \dots, K$ ), and  $M$  dependent variables, denoted as  $\mathbf{y}_m$  (where  $m = 1, \dots, M$ ). The data is structured into two matrices:  $\mathbf{X}$  with dimensions  $(N \times K)$  and  $\mathbf{Y}$  with dimensions  $(N \times M)$ .

After the model is constructed, it can be used to make predictions for new observations based on their  $\mathbf{X}$ -values. The output includes predicted  $\mathbf{X}$ -scores, residuals for the  $\mathbf{X}$  variables, the standard deviations of these residuals, and predicted  $\mathbf{Y}$ -values with associated confidence intervals (Wold et al., 2001).

According to lecture materials, the model predictions  $\hat{\mathbf{Y}}$  can be expressed in a following way shown in Equation 2. (Reinikainen, 2025)

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{W}(\mathbf{P}^T\mathbf{W})^{-1}\mathbf{Q}^T) \quad (2)$$

where  $\mathbf{X}$  is input data,  $\mathbf{W}$  is weight matrix,  $\mathbf{P}$  is orthogonal loading matrix and  $\mathbf{Q}$  is orthogonal loading matrix.

### 3. Results and discussion

The modeling goal was to reduce the model using only sensors that contribute most in terms of predicting the RUL of the engine. The main file consists of a loop in which all main phases are executed for each data frame.

During the calibration phase, the model complexity was optimized using an expanding-window cross-validation approach applied to the calibration subset. An analysis using PRESS and  $Q^2_{cv}$  scores was conducted to find the optimal number of components to use.

The model was then retrained using the optimal number of LVs, and its performance was evaluated on an independent validation set and then on the final test set.

#### 3.1. FD001

Optimal number of components based on  $Q^2_{cv}$ : 5

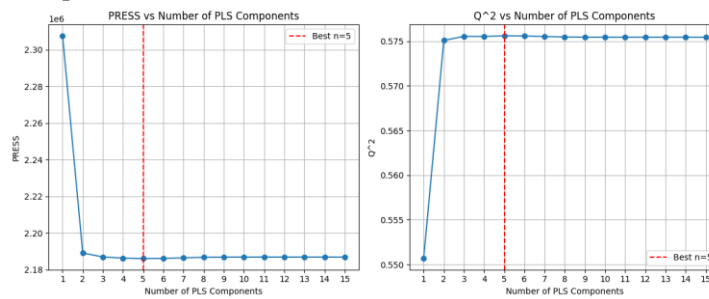


Figure 7. PRESS vs number of PLS components FD001

Selected sensors (5):

- sensor measurement 6,
- sensor measurement 3,
- sensor measurement 17,
- sensor measurement 11,
- sensor measurement 8

Table 1. Validation and testing metrics for FD001

|              | Validation   | Testing     |
|--------------|--------------|-------------|
| PRESS        | 13100244.783 | 7611107.759 |
| RMSE         | 54.124       | 43.577      |
| $Q^2_{test}$ | 0.514        | 0.564       |

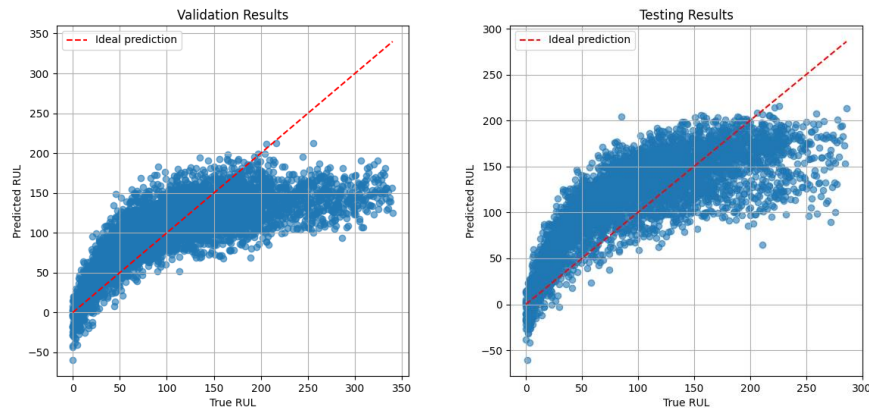


Figure 8. FD001 results

### 3.2. FD002

Optimal number of components based on  $Q^2_{cv}$ : 15.

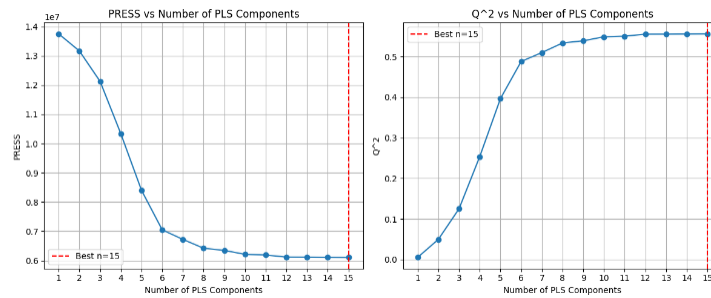


Figure 9. PRESS vs number of PLS components FD002

The  $Q^2_{cv}$  value kept increasing up to the maximum tested number of components (15), indicating that no clear optimum was reached. This behavior suggests potential overfitting, as the model continues to improve its fit on calibration data without necessarily improving generalization.

Selected sensors (7):

- sensor measurement 16,
- sensor measurement 11,
- sensor measurement 4,
- sensor measurement 15,
- sensor measurement 14,
- sensor measurement 17,
- sensor measurement 3

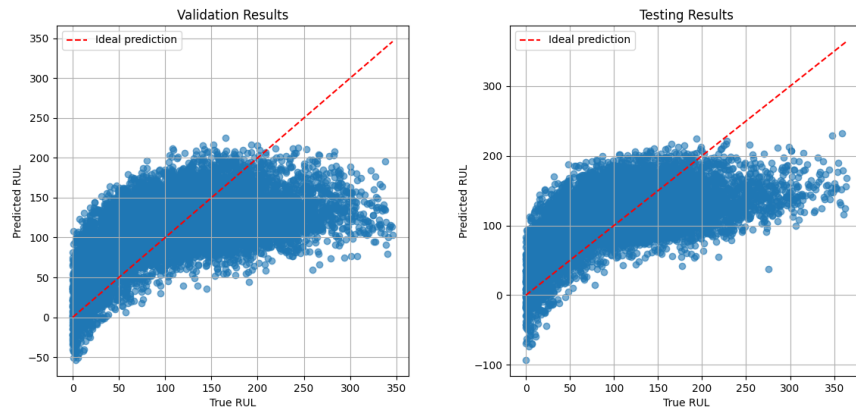


Figure 10. FD002 results

Table 2. Validation and testing metrics for FD002

|              | Validation   | Testing      |
|--------------|--------------|--------------|
| $PRESS$      | 36251770.542 | 29284833.137 |
| $RMSE$       | 56.548       | 52.505       |
| $Q^2_{test}$ | 0.381        | 0.411        |

### 3.3. FD003

Optimal number of components based on  $Q^2_{cv}$ : 2

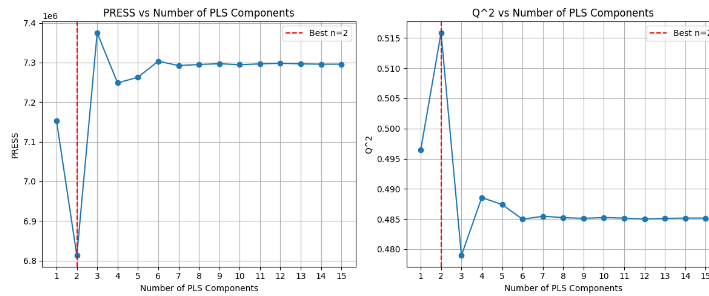


Figure 11. PRESS vs number of PLS components FD003

Selected sensors (6):

- sensor measurement 21,
- sensor measurement 20,
- sensor measurement 15,
- sensor measurement 7,
- sensor measurement 12,
- sensor measurement 10

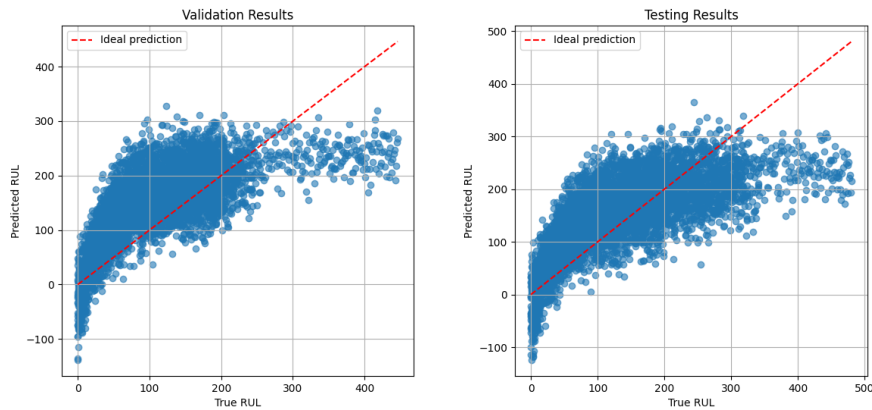


Figure 32. FD003 results

Table 3. Validation and testing metrics for FD003

|              | Validation   | Testing      |
|--------------|--------------|--------------|
| $PRESS$      | 21506602.355 | 23853647.850 |
| $RMSE$       | 70.435       | 68.070       |
| $Q^2_{test}$ | 0.319        | 0.484        |

### 3.4. FD004

Optimal number of components based on  $Q^2_{cv}$ : 15

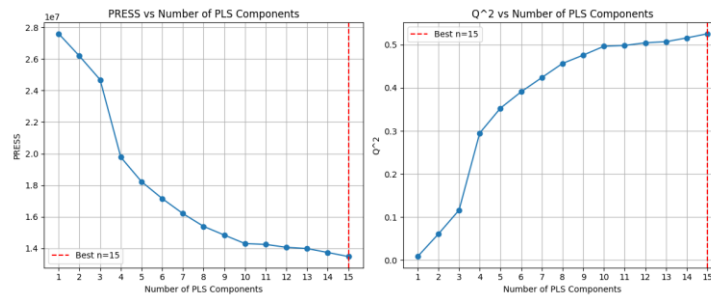


Figure 43. PRESS vs number of PLS components FD004

The final model achieved a low  $Q^2_{test}$  value ( $\sim 0.011$ ), suggesting limited generalization ability. This may result from the high operational variability in FD004 and the linear nature of the PLS model, which may not capture nonlinear dependencies between sensors and RUL.

Selected sensors (4):

- sensor measurement 14,
- sensor measurement 16,
- sensor measurement 11,
- sensor measurement 4

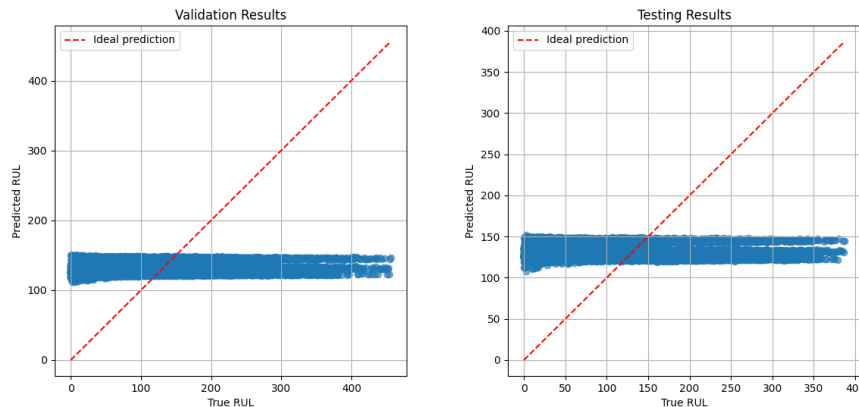


Figure 54. FD004 results

Table 4. Validation and testing metrics for FD004

|              | Validation    | Testing      |
|--------------|---------------|--------------|
| <i>PRESS</i> | 103929123.214 | 88026386.490 |
| <i>RMSE</i>  | 91.679        | 84.950       |
| $Q^2_{test}$ | 0.011         | 0.006        |

## 4. Conclusions

The models calibrated on the first dataframes show good predictive performance, and most sensor measurements were removed, thereby reducing the complexity of the model as asked in the task, but improvements can be made. However, as seen in Table 4, the  $Q^2$  calculated from the fourth dataframe is almost zero, so in this case the model cannot explain any variance and is not a good fit.

This suggests that the calibration part was not very successful, and maybe the general method can be modified, or other cross validation methods can be implemented. Also, it could be suggesting that the relationship between X and Y is not linear, since PLS assumes a linear relationship. The correlation coefficient  $r^2$  can be computed to assess this statement. A k-PLS or another non-linear method can be implemented to see if the model has a better RUL estimation.

## 5. References

Esposito Vinzi, Vincenzo & Chin, Wynne & Henseler, Jorg & Wang, Huiwen & St, Cahyono. (2022). Handbook of Partial Least Squares Concepts, Methods and Applications.

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